

## PROCEDURE

```
mysql> DELIMITER //
```

```
mysql>
```

```
mysql> CREATE PROCEDURE GetSquare(IN num INT)
```

```
    -> BEGIN
```

```
        -> SELECT num * num AS Square;
```

```
    -> END //
```

```
Query OK, 0 rows affected (0.02 sec)
```

  

```
mysql>
```

```
mysql> DELIMITER ;
```

```
mysql>
```

```
mysql> CALL GetSquare(5);
```

| Square |
|--------|
| 25     |

```
1 row in set (0.00 sec)
```

## PROCEDURE WITH IF-ELSE

```
mysql> DELIMITER //
```

```
mysql>
```

```
mysql> CREATE FUNCTION GetGrade(mark INT)
```

```
    -> RETURNS VARCHAR(10)
```

```
    -> DETERMINISTIC
```

```
    -> BEGIN
```

```
        -> IF mark >= 90 THEN
```

```
            -> RETURN 'A';
```

```
        -> ELSEIF mark >= 75 THEN
```

```
            -> RETURN 'B';
```

```
        -> ELSEIF mark >= 50 THEN
```

```
            -> RETURN 'C';
```

```
        -> ELSE
```

```
            -> RETURN 'Fail';
```

```
        -> END IF;
```

```
    -> END //
```

```
Query OK, 0 rows affected (0.01 sec)
```

## PROCEDURE WITH CASE

```
mysql> DELIMITER //
```

```
mysql>
```

```
mysql> CREATE PROCEDURE CheckGrade(IN mark INT)
```

```
  -> BEGIN
```

```
  ->     CASE
```

```
  ->         WHEN mark >= 90 THEN
```

```
  ->             SELECT 'A' AS Grade;
```

```
  ->         WHEN mark >= 75 THEN
```

```
  ->             SELECT 'B' AS Grade;
```

```
  ->         WHEN mark >= 50 THEN
```

```
  ->             SELECT 'C' AS Grade;
```

```
  ->         ELSE
```

```
  ->             SELECT 'Fail' AS Grade;
```

```
  ->     END CASE;
```

```
  -> END //
```

```
Query OK, 0 rows affected (0.01 sec)
```

```
mysql>
```

```
mysql> DELIMITER ;
```

```
mysql>
```

```
mysql> CALL CheckGrade(85);
```

| Grade |
|-------|
| B     |

```
1 row in set (0.00 sec)
```

## PROCEDURE USING LOOP

```
mysql> DELIMITER //
```

```
mysql>
```

```
mysql> CREATE PROCEDURE NumberLoop()
```

```
  -> BEGIN
```

```
  ->     DECLARE i INT DEFAULT 1;
```

```
  ->
```

```
  ->     loop1: LOOP
```

```
  ->         SELECT i;
```

```
  ->
```

```
  ->         SET i = i + 1;
```

```
  ->
```

```
  ->         IF i > 5 THEN
```

```
  ->             LEAVE loop1;
```

```
  ->         END IF;
```

```
  ->     END LOOP;
```

```
  -> END //
```

```
Query OK, 0 rows affected (0.02 sec)
```

```
mysql>
```

```
mysql> DELIMITER ;
```

```
mysql>
```

```
mysql> CALL NumberLoop();
```

| i |
|---|
| 1 |

```
1 row in set (0.00 sec)
```

  

| i |
|---|
| 2 |

```

+-----+
| i      |
+-----+
|      3 |
+-----+
1 row in set (0.02 sec)

+-----+
| i      |
+-----+
|      4 |
+-----+
1 row in set (0.02 sec)

+-----+
| i      |
+-----+
|      5 |
+-----+
1 row in set (0.04 sec)

```

#### PROCEDURE USING WHILE LOOP

```

mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE NumberWhile()
-> BEGIN
->     DECLARE i INT DEFAULT 1;
->
->     WHILE i <= 5 DO
->         SELECT i;
->         SET i = i + 1;
->     END WHILE;
-> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql>
mysql> CALL NumberWhile();
+-----+
| i      |
+-----+
|      1 |
+-----+
1 row in set (0.00 sec)

+-----+
| i      |
+-----+
|      2 |
+-----+
1 row in set (0.01 sec)

```

```

+-----+
| i      |
+-----+
|      3 |
+-----+
1 row in set (0.03 sec)

+-----+
| i      |
+-----+
|      4 |
+-----+
1 row in set (0.03 sec)

+-----+
| i      |
+-----+
|      5 |
+-----+
1 row in set (0.04 sec)

Query OK, 0 rows affected (0.04 sec)

```

## PROCEDURE USING REPEAT LOOP

```

mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE NumberRepeat()
-> BEGIN
->     DECLARE i INT DEFAULT 1;
->
->     REPEAT
->         SELECT i;
->         SET i = i + 1;
->     UNTIL i > 5
->     END REPEAT;
-> END //
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> DELIMITER ;
mysql>
mysql> CALL NumberRepeat();
+-----+
| i      |
+-----+
|      1 |
+-----+
1 row in set (0.00 sec)

+-----+
| i      |
+-----+
|      2 |
+-----+
1 row in set (0.01 sec)

```

```

+-----+
| i      |
+-----+
|      3 |
+-----+
1 row in set (0.03 sec)

+-----+
| i      |
+-----+
|      4 |
+-----+
1 row in set (0.03 sec)

+-----+
| i      |
+-----+
|      5 |
+-----+
1 row in set (0.04 sec)

Query OK, 0 rows affected (0.04 sec)

```

#### DELETE A PROCEDURE

```

mysql> DROP PROCEDURE IF EXISTS GetSquare;
Query OK, 0 rows affected (0.02 sec)

```

#### FUNCTION

```

mysql> use 22a;
Database changed
mysql> DELIMITER //
mysql>
mysql> CREATE FUNCTION GetSquare(num INT)
-> RETURNS INT
-> DETERMINISTIC
-> BEGIN
->     RETURN num * num;
-> END //
Query OK, 0 rows affected (0.06 sec)

mysql>
mysql> DELIMITER ;
mysql>
mysql> SELECT GetSquare(5);
+-----+
| GetSquare(5) |
+-----+
|           25 |
+-----+
1 row in set (0.01 sec)

```

## IF-THEN-ELSE FUNCTION

```
mysql> DELIMITER //
mysql>
mysql> CREATE FUNCTION CheckNumber(num INT)
  -> RETURNS VARCHAR(20)
  -> DETERMINISTIC
  -> BEGIN
  ->     IF num >= 0 THEN
  ->         RETURN 'Positive';
  ->     ELSE
  ->         RETURN 'Negative';
  ->     END IF;
  -> END //
```

Query OK, 0 rows affected (0.01 sec)

```
mysql>
mysql> DELIMITER ;
mysql>
mysql> SELECT CheckNumber(10);
```

```
+-----+
| CheckNumber(10) |
+-----+
| Positive        |
+-----+
1 row in set (0.00 sec)
```

## IF-ELSEIF-ELSE FUNCTION

```
mysql> DELIMITER //
mysql>
mysql> CREATE FUNCTION GetGrade(mark INT)
  -> RETURNS VARCHAR(10)
  -> DETERMINISTIC
  -> BEGIN
  ->     IF mark >= 90 THEN
  ->         RETURN 'A';
  ->     ELSEIF mark >= 75 THEN
  ->         RETURN 'B';
  ->     ELSEIF mark >= 50 THEN
  ->         RETURN 'C';
  ->     ELSE
  ->         RETURN 'Fail';
  ->     END IF;
  -> END //
```

Query OK, 0 rows affected (0.01 sec)

```
mysql>
mysql> DELIMITER ;
mysql>
mysql> SELECT GetGrade(85);
```

```
+-----+
| GetGrade(85) |
+-----+
| B            |
+-----+
1 row in set (0.00 sec)
```

## CASE STATEMENT

```
mysql> DELIMITER //
```

```
mysql>
```

```
mysql> CREATE FUNCTION Grade(mark INT)
```

```
    -> RETURNS VARCHAR(10)
```

```
    -> DETERMINISTIC
```

```
    -> BEGIN
```

```
    ->     DECLARE result VARCHAR(10);
```

```
    ->
```

```
    ->     CASE
```

```
    ->         WHEN mark >= 90 THEN
```

```
    ->             SET result = 'A';
```

```
    ->         WHEN mark >= 75 THEN
```

```
    ->             SET result = 'B';
```

```
    ->         WHEN mark >= 50 THEN
```

```
    ->             SET result = 'C';
```

```
    ->         ELSE
```

```
    ->             SET result = 'FAIL';
```

```
    ->     END CASE;
```

```
    ->
```

```
    ->     RETURN result;
```

```
    -> END //
```

Query OK, 0 rows affected (0.02 sec)

```
mysql>
```

```
mysql> DELIMITER ;
```

```
mysql>
```

```
mysql> SELECT Grade(85);
```

| Grade(85) |
|-----------|
| B         |

```
1 row in set (0.00 sec)
```

## LOOP

```
mysql> DELIMITER //
mysql>
mysql> CREATE FUNCTION SumLoop(n INT)
-> RETURNS INT
-> DETERMINISTIC
-> BEGIN
->     DECLARE i INT DEFAULT 1;
->     DECLARE total INT DEFAULT 0;
->
->     loop1: LOOP
->         SET total = total + i;
->         SET i = i + 1;
->
->         IF i > n THEN
->             LEAVE loop1;
->         END IF;
->     END LOOP;
->
->     RETURN total;
-> END //
```

Query OK, 0 rows affected (0.01 sec)

```
mysql>
mysql> DELIMITER ;
mysql>
mysql> SELECT SumLoop(5);
+-----+
| SumLoop(5) |
+-----+
|          15 |
+-----+
1 row in set (0.00 sec)
```



## WHILE LOOP

```
mysql> DELIMITER //
mysql>
mysql> CREATE FUNCTION SumWhile(n INT)
-> RETURNS INT
-> DETERMINISTIC
-> BEGIN
->     DECLARE i INT DEFAULT 1;
->     DECLARE total INT DEFAULT 0;
->
->     WHILE i <= n DO
->         SET total = total + i;
->         SET i = i + 1;
->     END WHILE;
->
->     RETURN total;
-> END //
```

Query OK, 0 rows affected (0.02 sec)

```
mysql>
mysql> DELIMITER ;
mysql>
mysql> SELECT SumWhile(5);
+-----+
| SumWhile(5) |
+-----+
|          15 |
+-----+
1 row in set (0.00 sec)
```

## REPEAT LOOP

```
mysql> DELIMITER //
mysql>
mysql> CREATE FUNCTION SumRepeat(n INT)
-> RETURNS INT
-> DETERMINISTIC
-> BEGIN
->     DECLARE i INT DEFAULT 1;
->     DECLARE total INT DEFAULT 0;
->
->     REPEAT
->         SET total = total + i;
->         SET i = i + 1;
->     UNTIL i > n
->     END REPEAT;
->
->     RETURN total;
-> END //
```

Query OK, 0 rows affected (0.02 sec)

```
mysql>
mysql> DELIMITER ;
mysql>
mysql> SELECT SumRepeat(5);
+-----+
| SumRepeat(5) |
+-----+
|          15 |
+-----+
1 row in set (0.00 sec)
```